

COLLEGE EXAMINATIONS – ACADEMIC YEAR 2017- 2018**SCHOOL OF SCIENCE****DEPARTMENT OF APPLIED MATHEMATICS****FIRST YEAR: CIV-CE, -SGE, -SE, -W, -EE, MEE, CS, ETE, CSE, ELE, IS, IT, CMT, Geology and Mining****SEMESTER II****FINAL EXAMINATION****MATHEMATICS FOR ENGINEERERS II (MAT1264)****DATE:/May/2018****TIME: 2 hours****MAXIMUM MARKS = 50****INSTRUCTIONS:**

1. This paper contains **TWO** sections.
 2. Section A is compulsory, and Answer any **TWO** of the **THREE** questions in Section B.
 3. No written materials allowed into the Examination Room.
 4. Write all your answers in the answer booklet provided.
 5. Do not forget to write your Registration Number.
 6. Do not write any answers on this question paper.
 7. Where appropriate draw large clearly labeled diagrams in your answers.
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Section A

Question 1 (20 marks)

1. (a) Solve the linear differential equation

$$x \frac{dy}{dx} + (x+1)y = 2. \quad (3 \text{ marks})$$

- (b) Find the general solution of the following differential equation

$$y'' - 2y' - 3y = 5 \cos(2t). \quad (3 \text{ marks})$$

- (c) Calculate the Laplace transform of the function $f(t) = t \sin(3t)$. (2 marks)

- (d) Consider $\vec{F} = (x^2 - y)\vec{i} + 4z\vec{j} + x^2\vec{k}$. Find $\vec{\nabla} \times \vec{F}$. (2 marks)

- (e) Determine if the set S of functions $\{1, e^x, e^{-x}\}$ is linearly independent or dependent using the Wronskian. (3 marks)

- (f) Indicate (sketch) the region of integration R and hence evaluate $\iint_R (x^2 + y^2) dy dx$ where R is the triangle formed by the lines $y = 0$, $y = x$ and $x = 2$. (3 marks)

- (g) i. State the **Kirchhoff's Voltage Law**. (1 mark)
 ii. An RCL circuit connected in series has $R=180$ ohms, $C=1/280$ farad, $L=20$ henries, and an applied voltage $E(t)=10 \sin(t)$. Assuming no initial charge on the capacitor, but an initial current of 1 ampere at $t=0$ when the voltage is first applied, find the subsequent charge on the capacitor. (3 marks)

Section B

Question 2 (15 marks)

2. (a) Solve the system $x'(t) = \begin{bmatrix} 3 & -2 \\ 2 & -2 \end{bmatrix} x(t)$ with $x(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ using Eigenvalue method. (7 marks)

- (b) Consider the vector Field $\vec{F} = (2xy)\vec{i} + (x^2 + z^3)\vec{j} + (3yz + 4z^3)\vec{k}$. Show that $\vec{F}$ is conservative i.e $\vec{F} = \nabla f$ and hence, find the potential f . (5 marks)

- (c) Find the inverse Laplace transform of

$$F(s) = \frac{s+7}{s^2+2s+5} \quad (3 \text{ marks}).$$

Question 3 (15 marks)

3. (a) Evaluate

$$\iint_R \frac{e^y}{y} dy dx$$

where R is the region enclosed by $y = x$ and $y = \sqrt{x}$.

(6 marks)

(b) i. State the **Green's theorem** in the plane.

(2 marks)

ii. Use the theorem in 3.(b) i. to evaluate the following integral

$$\oint_C (y^2 - 7y) dx + (2xy + 2x) dy \quad \text{along the unit circle } C.$$

(4 marks)

(c) Solve Cauchy-Euler equation $x^2 y'' + 7xy' + 8y = 0$.

(3 marks)

Question 4 (15 marks)

4. (a) Define

i. An Ordinary Differential Equation (ODE);

(1 mark)

ii. Gradient of a scalar function.

(1 mark)

(b) Find the Jacobian of the transformation $u = x$ and $v = xy$.

(2 marks)

(c) Solve the differential equation $y'' - 4y' + 4y = \frac{e^{2x}}{x}$.

(5 marks)

(d) Evaluate

$$\iiint_G xyz dV$$

where G is the domain of $\mathbb{R}^3$ limited by $x = 0, y = 0, z = 0$, and $x + y + z = 1$

(6 marks)

All The Best!!